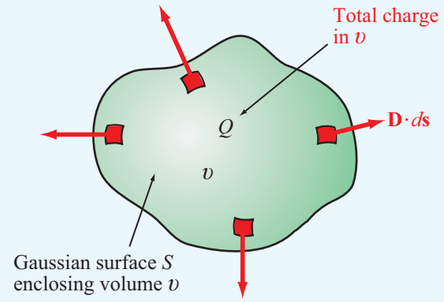


Figure 4-8 The integral form of Gauss's law states that the outward flux of **D** through a surface is proportional to the enclosed charge Q.

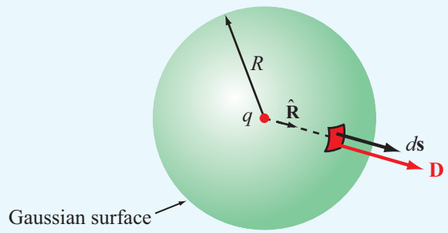


Figure 4-9 Electric field **D** due to point charge q.

and $d\mathbf{s} = \hat{\mathbf{R}}\, ds$. Applying Gauss's law gives

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = \oint_S \hat{\mathbf{R}} D_R \cdot \hat{\mathbf{R}}\, ds = \oint_S D_R\, ds = D_R(4\pi R^2) = q. \tag{4.31}$$

Solving for D_R and then inserting the result in Eq. (4.30) gives the expression for the electric field **E** induced by an isolated point charge in a medium with permittivity ε:

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon} = \hat{\mathbf{R}}\,\frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}. \tag{4.32}$$

This is identical with Eq. (4.13) obtained from Coulomb's law; after all, Maxwell's equations incorporate Coulomb's law. For this simple case of an isolated point charge, it does not matter whether Coulomb's law or Gauss's law is used to obtain the expression for **E**. However, it does matter which approach we follow when we deal with multiple point charges or continuous charge distributions. Even though Coulomb's law can be used to find **E** for any specified distribution of charge, Gauss's law is easier to apply than Coulomb's law, but its utility is limited to symmetrical charge distributions.

► Gauss's law, as given by Eq. (4.29), provides a convenient method for determining the flux density **D** when the charge distribution possesses symmetry properties that allow us to infer the variations of the magnitude and direction of **D** as a function of spatial location. This facilitates the integration of **D** over a cleverly chosen Gaussian surface. ◄

Because at every point on the surface the direction of $d\mathbf{s}$ is along its outward normal, only the normal component of **D** at the surface contributes to the integral in Eq. (4.29). To successfully apply Gauss's law, the surface S should be chosen from symmetry considerations so that, across each subsurface of S, **D** is constant in magnitude and its direction is either normal or purely tangential to the subsurface. These aspects are illustrated in Example 4-6.

Example 4-6: Electric Field of an Infinite Line Charge

Use Gauss's law to obtain an expression for **E** due to an infinitely long line with uniform charge density ρ_ℓ that resides along the z axis in free space.

Solution: Since the charge density along the line is uniform, infinite in extent, and residing along the z axis, symmetry considerations dictate that **D** is in the radial $\hat{\mathbf{r}}$ direction and cannot depend on ϕ or z. Thus, $\mathbf{D} = \hat{\mathbf{r}} D_r$. Therefore, we construct a finite cylindrical Gaussian surface of radius r and height h, which is concentric around the line of charge (Fig. 4-10). The total charge contained within the cylinder is $Q = \rho_\ell h$. Since **D** is along $\hat{\mathbf{r}}$, the top and bottom surfaces of the cylinder do not contribute to the surface integral on the left-hand side of Eq. (4.29); that is, only the curved surface contributes to the integral. Hence,

$$\int_{z=0}^{h}\int_{\phi=0}^{2\pi} \hat{\mathbf{r}} D_r \cdot \hat{\mathbf{r}} r\, d\phi\, dz = \rho_\ell h$$

or

$$2\pi h D_r r = \rho_\ell h,$$

which yields

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_0} = \hat{\mathbf{r}}\,\frac{D_r}{\varepsilon_0} = \hat{\mathbf{r}}\,\frac{\rho_\ell}{2\pi\varepsilon_0 r}. \tag{4.33}$$

(infinite line charge)

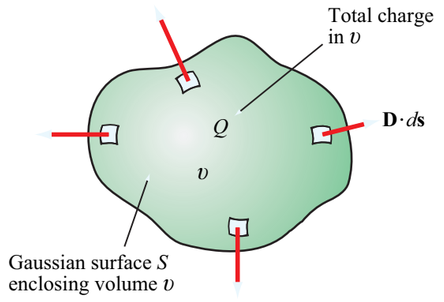


图 4-8 高斯定律的积分形式指出，通过表面的 D 的向外通量与所包含的电荷 Q 成正比。

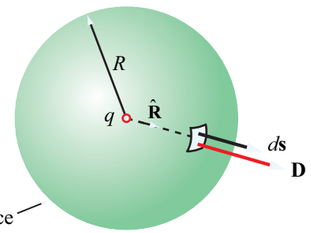


图 4-9 点电荷 q 产生的电场 D。

和 ds = R̂ ds。应用高斯定律给出

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = \oint_S \hat{\mathbf{R}} D_R \cdot \hat{\mathbf{R}}\, ds = \oint_S D_R\, ds = D_R(4\pi R^2) = q. \tag{4.31}$$

求解 D_R，然后将结果代入方程 (1)。(4.30) 给出了介电常数为 ε 的介质中孤立点电荷感应的电场 E 的表达式：

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon} = \hat{\mathbf{R}}\,\frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}. \tag{4.32}$$

这与等式相同。(4.13) 由库仑定律获得；毕竟，麦克斯韦方程组包含了库仑定律。对于孤立点电荷的这种简单情况，使用库仑定律还是高斯定律来获得 E 的表达式并不重要。但是，当我们处理多个点电荷或连续电荷分布时，我们采用哪种方法就很重要了。尽管可以使用库仑定律

对于任何指定的电荷分布，要求 E，高斯定律比库仑定律更容易应用，但其实用性仅限于对称电荷分布。

► 高斯定律，如方程所示。（4.29）提供了一种在电荷分布具有对称性时确定通量密度 D 的简便方法，使我们能够推断出 D 的大小和方向随空间位置的函数的变化。这有利于 D 在巧妙选择的高斯表面上的积分。◄

因为在表面上的每个点，ds 的方向都是沿着其向外法线的，因此只有表面上 D 的法线分量对方程 (1) 中的积分有贡献。（4.29）。为了成功应用高斯定律，应根据对称性考虑选择表面 S，以便在 S 的每个子表面上，D 的大小恒定，并且其方向与子表面垂直或纯相切。这些方面在示例 4-6 中进行了说明。

例 4-6：无限线电荷的电场

由于自由空间中沿 z 轴存在一条具有均匀电荷密度 ρ_ℓ 的无限长线，因此可以使用高斯定律获得 E 的表达式。

解：由于沿线的电荷密度是均匀的、无限大的且位于 z 轴上，对称性考虑表明 D 位于径向 r̂ 方向，并且不能依赖于 ϕ 或 z。因此，D = r̂ D_r。因此，我们构造一个半径为 r、高度为 h 的有限圆柱高斯表面，它与电荷线同心（图 4-10）。气缸内的总充气量为 Q = ρ_ℓh。由于 D 沿 r̂，因此圆柱体的顶面和底面对方程左侧的表面积分没有贡献。（4.29）；也就是说，只有曲面对积分有贡献。因此，

$$\int_{z=0}^{h}\int_{\phi=0}^{2\pi} \hat{\mathbf{r}} D_r \cdot \hat{\mathbf{r}} r\, d\phi\, dz = \rho_\ell h$$

or

$$2\pi h D_r r = \rho_\ell h,$$

这产生

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_0} = \hat{\mathbf{r}}\,\frac{D_r}{\varepsilon_0} = \hat{\mathbf{r}}\,\frac{\rho_\ell}{2\pi\varepsilon_0 r}. \tag{4.33}$$

(infinite line charge)

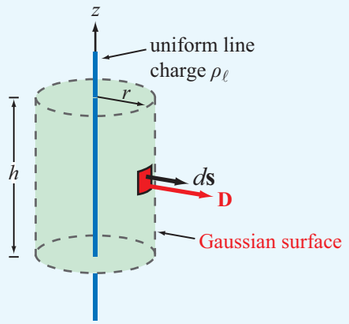


Figure 4-10 Gaussian surface around an infinitely long line of charge (Example 4-6).

Note that Eq. (4.33) is applicable for any infinite line of charge, regardless of its location and direction, as long as $\hat{\mathbf{r}}$ is properly defined as the radial distance vector from the line charge to the observation point (i.e., $\hat{\mathbf{r}}$ is perpendicular to the line charge to the observation point).

Example 4-7: Two Infinite Lines of Charge

Figure 4-11 depicts the presence of two infinite lines of charge in free space: one residing in the x–y plane parallel to the $\hat{\mathbf{x}}$ axis and carrying charge density $\rho_{\ell_1} = 1$ (nC/m), and a second one residing in the y–z plane parallel to the y axis and carrying a charge density $\rho_{\ell_2} = -2$ (nC/m). Determine the electric field at the origin.

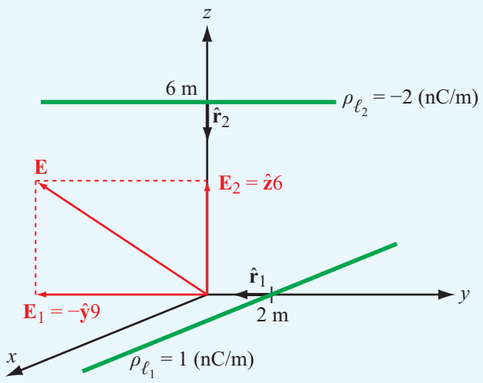


Figure 4-11 Two infinite lines of charge (Example 4-7).

Solution: The electric field $\mathbf{E}$ is the sum of two electric field components:

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2,$$

where $\mathbf{E}_1$ and $\mathbf{E}_2$ are the electric fields due to line charges 1 and 2, respectively. According to Eq. (4.33), the direction of the electric field $\hat{\mathbf{r}}$ is perpendicular to the direction of the line charge and points away from the line of charge (if ρ_ℓ is positive). Hence, for the first line of charge, $\rho_{\ell_1} = 1$ (nC/m), $\hat{\mathbf{r}}_1 = -\hat{\mathbf{y}}$, $r_1 = 2$, and

$$\mathbf{E}_1 = \frac{\hat{\mathbf{r}}_1 \rho_{\ell_1}}{2\pi\varepsilon_0 r_1} = \frac{-\hat{\mathbf{y}}\, 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 2} = -\hat{\mathbf{y}}\, 9 \quad \text{V/m.}$$

Similarly, for the second line of charge, $\rho_{\ell_2} = -2$ (nC/m), $\hat{\mathbf{r}}_2 = -\hat{\mathbf{z}}$, $r_2 = 6$, and

$$\mathbf{E}_2 = \frac{\hat{\mathbf{r}}_2 \rho_{\ell_2}}{2\pi\varepsilon_0 r_2} = \frac{-\hat{\mathbf{z}}(-2) \times 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 6} = \hat{\mathbf{z}}\, 6 \quad \text{V/m.}$$

Hence,

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = (-\hat{\mathbf{y}}\, 9 + \hat{\mathbf{z}}\, 6) \quad \text{V/m.}$$

Concept Question 4-7: Explain Gauss's law. Under what circumstances is it useful?

Concept Question 4-8: How should one choose a Gaussian surface?

Exercise 4-7: Two infinite lines, each carrying a uniform charge density ρ_ℓ, reside in free space parallel to the z axis at $x = 1$ and $x = -1$. Determine $\mathbf{E}$ at an arbitrary point along the y axis.

Answer: $\mathbf{E} = \hat{\mathbf{y}}\rho_\ell y / \left[\pi\varepsilon_0(y^2+1)\right]$. (See EM.)

Exercise 4-8: A thin spherical shell of radius a carries a uniform surface charge density ρ_s. Use Gauss's law to determine $\mathbf{E}$ everywhere in free space.

Answer: $\mathbf{E} = \begin{cases} 0 & \text{for } R < a; \\ \hat{\mathbf{R}}\rho_s a^2/(\varepsilon R^2) & \text{for } R > a. \end{cases}$ (See EM.)

182 CHAPTER 4 ELECTROSTATICS

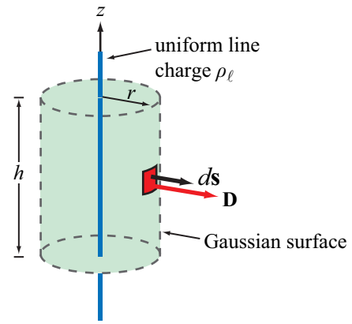


图 4-10 无限长电荷线周围的高斯表面（示例 4-6）。

请注意等式。只要将 $\hat{\mathbf{r}}$ 正确定义为从线电荷到观察点的径向距离矢量（即 $\hat{\mathbf{r}}$ 垂直于电荷线），方程（4.33）适用于任何无限电荷线，无论其位置和方向如何。

例 4-7：两条无限电荷线

图 4-11 描述了自由空间中存在两条无限电荷线：一条位于平行于 $\hat{\mathbf{x}}$ 轴的 x-y 平面中

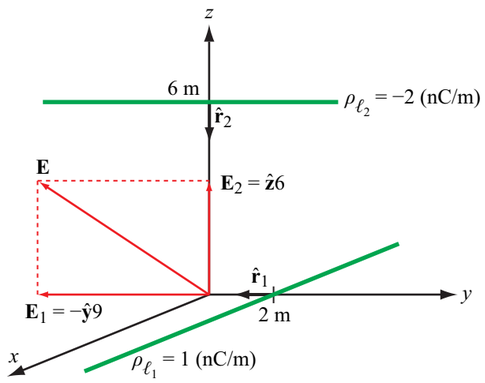


图 4-11 两条无限电荷线（例 4-7）。

并携带电荷密度 $\rho_{\ell_1} = 1$ (nC/m)，第二个位于平行于 y 轴的 y-z 平面并携带电荷密度 $\rho_{\ell_2} = -2$ (nC/m)。确定原点处的电场。

解：电场 E 是两个电场分量之和：

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2,$$

其中 E_1 和 E_2 分别是线电荷 1 和 2 产生的电场。根据方程。（4.33）中，电场 $\hat{\mathbf{r}}$ 的方向垂直于线电荷的方向，并且指向远离电荷线的方向（如果 ρ_ℓ 为正）。因此，对于第一条电荷线，$\rho_{\ell_1} = 1$ (nC/m)、$\hat{\mathbf{r}}_1 = -\hat{\mathbf{y}}$、$r_1 = 2$ 和

$$\mathbf{E}_1 = \frac{\hat{\mathbf{r}}_1 \rho_{\ell_1}}{2\pi\varepsilon_0 r_1} = \frac{-\hat{\mathbf{y}}\, 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 2} = -\hat{\mathbf{y}}\, 9 \quad \text{V/m.}$$

同样，对于第二条电荷线，$\rho_{\ell_2} = -2$ (nC/m)、$\hat{\mathbf{r}}_2 = -\hat{\mathbf{z}}$、$r_2 = 6$ 和

$$\mathbf{E}_2 = \frac{\hat{\mathbf{r}}_2 \rho_{\ell_2}}{2\pi\varepsilon_0 r_2} = \frac{-\hat{\mathbf{z}}(-2) \times 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 6} = \hat{\mathbf{z}}\, 6 \quad \text{V/m.}$$

因此，

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = (-\hat{\mathbf{y}}\, 9 + \hat{\mathbf{z}}\, 6) \quad \text{V/m.}$$

概念问题 4-7：解释高斯定律。什么情况下有用？

概念题4-8：应该如何选择高斯曲面？

练习 4-7：两条无限长的线，每条都带有均匀的电荷密度 ρ_ℓ，位于平行于 z 轴的自由空间中 $x = 1$ 和 $x = -1$ 处。确定沿 y 轴任意点的 E。

Answer: $\mathbf{E} = \hat{\mathbf{y}}\rho_\ell y / \left[\pi\varepsilon_0(y^2+1)\right]$. (See EM.)

练习 4-8：半径为 a 的薄球壳具有均匀的表面电荷密度 ρ_s。使用高斯定律确定自由空间中各处的 E。

Answer: $\mathbf{E} = \begin{cases} 0 & \text{for } R < a; \\ \hat{\mathbf{R}}\rho_s a^2/(\varepsilon R^2) & \text{for } R > a. \end{cases}$ (See EM.)

Exercise 4-9: A spherical volume of radius a contains a uniform volume charge density ρ_v. Use Gauss's law to determine $\mathbf{D}$ for (a) $R \le a$ and (b) $R \ge a$.

Answer: (a) $\mathbf{D} = \hat{\mathbf{R}}\rho_v R/3$,
(b) $\mathbf{D} = \hat{\mathbf{R}}\rho_v a^3/(3R^2)$. (See EM.)

4-5 Electric Scalar Potential

The operation of an electric circuit usually is described in terms of the currents flowing through its branches and the voltages at its nodes. The voltage difference V between two points in a circuit represents the amount of work, or *potential* energy, required to move a unit charge from one to the other.

► The term "voltage" is short for "voltage potential" and synonymous with ***electric potential***. ◄

Even though when analyzing a circuit we may not consider the electric fields present in the circuit, it is in fact the existence of these fields that gives rise to voltage differences across circuit elements such as resistors or capacitors. The relationship between the electric field $\mathbf{E}$ and the electric potential V is the subject of this section.

4-5.1 Electric Potential as a Function of Electric Field

We begin by considering the simple case of a positive charge q in a uniform electric field $\mathbf{E} = -\hat{\mathbf{y}}E$ in the $-y$ direction (**Fig. 4-12**). The presence of the field $\mathbf{E}$ exerts a force $\mathbf{F}_e = q\mathbf{E}$ on the charge in the $-y$ direction. To move the charge along the positive y direction (against the force $\mathbf{F}_e$), we need to provide an external force $\mathbf{F}_{ext}$ to counteract $\mathbf{F}_e$, which requires the expenditure of energy. To move q without acceleration (at constant speed), the net force acting on the charge must be zero, which means that $\mathbf{F}_{ext} + \mathbf{F}_e = 0$, or

$$\mathbf{F}_{ext} = -\mathbf{F}_e = -q\mathbf{E}. \tag{4.34}$$

The work done (or energy expended) in moving any object a vector differential distance $d\mathbf{l}$ while exerting a force $\mathbf{F}_{ext}$ is

$$dW = \mathbf{F}_{ext} \cdot d\mathbf{l} = -q\mathbf{E} \cdot d\mathbf{l} \quad \text{(J)}. \tag{4.35}$$

Work (or energy) is measured in joules (J). If the charge is moved a distance dy along $\hat{\mathbf{y}}$, then

$$dW = -q(-\hat{\mathbf{y}}E) \cdot \hat{\mathbf{y}}\, dy = qE\, dy. \tag{4.36}$$

The differential electric potential energy dW per unit charge is called the ***differential electric potential*** (or differential voltage) dV. That is,

$$dV = \frac{dW}{q} = -\mathbf{E} \cdot d\mathbf{l} \quad \text{(J/C or V)}. \tag{4.37}$$

The unit of V is the volt (V) with 1 V = 1 J/C, and since V is measured in volts, the electric field is expressed in volts per meter (V/m).

The ***potential difference*** corresponding to moving a point charge from point P_1 to point P_2 (**Fig. 4-13**) is obtained by integrating Eq. (4.37) *along any path* between them. That is,

$$\int_{P_1}^{P_2} dV = -\int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \tag{4.38}$$

or

$$V_{21} = V_2 - V_1 = -\int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \tag{4.39}$$

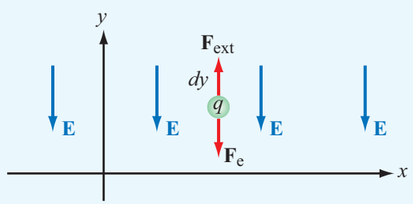


Figure 4-12 Work done in moving a charge q a distance dy against the electric field $\mathbf{E}$ is $dW = qE\, dy$.

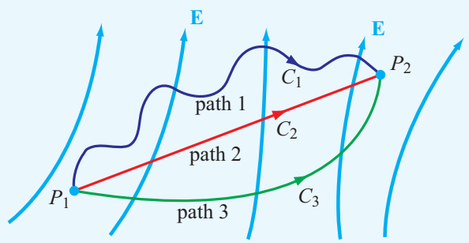


Figure 4-13 In electrostatics, the potential difference between P_2 and P_1 is the same irrespective of the path used for calculating the line integral of the electric field between them.

练习 4-9：半径为 a 的球形体积包含均匀的体积电荷密度 ρ_v。使用高斯定律确定 (a) $R \le a$ 和 (b) $R \ge a$ 的 D。

Answer: (a) $\mathbf{D} = \hat{\mathbf{R}}\rho_v R/3$,
(b) $\mathbf{D} = \hat{\mathbf{R}}\rho_v a^3/(3R^2)$. (See EM.)

4-5 电标量势

电路的运行通常用流经其支路的电流和其节点的电压来描述。电路中两点之间的电压差 V 表示将单位电荷从一个点移动到另一个点所需的功量或势能。

► 术语“电压”是“电压电势”的缩写，与电势同义。◄

尽管在分析电路时我们可能不会考虑电路中存在的电场，但实际上这些电场的存在会在电阻器或电容器等电路元件上产生电压差。电场 E 和电势 V 之间的关系是本节的主题。

4-5.1 电势作为电场的函数

我们首先考虑在 −y 方向均匀电场 $\mathbf{E} = -\hat{\mathbf{y}}E$ 中正电荷 q 的简单情况（图 4-12）。场 E 的存在在 −y 方向上对电荷施加力 $F_e = qE$。为了沿着正 y 方向移动电荷（对抗力 F_e），我们需要提供一个外力 F_{ext} 来抵消 F_e，需要消耗能量。要在没有加速的情况下（以恒定速度）移动 q，作用在电荷上的净力必须为零，这意味着 $F_{ext} + F_e = 0$，或

$$\mathbf{F}_{ext} = -\mathbf{F}_e = -q\mathbf{E}. \tag{4.34}$$

在施加力 F_{ext} 的同时将任何物体移动矢量差距离 dl 所做的功（或消耗的能量）为

$$dW = \mathbf{F}_{ext} \cdot d\mathbf{l} = -q\mathbf{E} \cdot d\mathbf{l} \quad \text{(J)}. \tag{4.35}$$

功（或能量）以焦耳 (J) 为单位。如果电荷沿ŷ移动距离dy，则

$$dW = -q(-\hat{\mathbf{y}}E) \cdot \hat{\mathbf{y}}\, dy = qE\, dy. \tag{4.36}$$

每单位电荷的微分电势能 dW 称为微分电势（或微分电压）dV 。那是，

$$dV = \frac{dW}{q} = -\mathbf{E} \cdot d\mathbf{l} \quad \text{(J/C or V)}. \tag{4.37}$$

V 的单位是伏特 (V)，其中 1 V = 1 J/C，由于 V 以伏特为单位测量，因此电场以伏特每米 (V/m) 表示。

图 1 点电荷从 P_1 点移动到 P_2（点对应的电势差4-13）是通过积分方程得到的。（4.37）沿着它们之间的任何路径。那是，

$$\int_{P_1}^{P_2} dV = -\int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \tag{4.38}$$

或者

$$V_{21} = V_2 - V_1 = -\int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \tag{4.39}$$

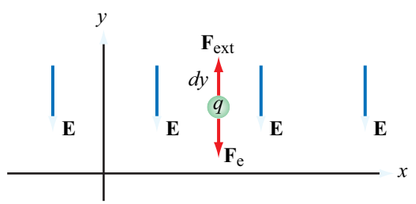


图 4-12 电荷 q 相对于电场 E 移动距离 dy 所做的功是 dW = qE dy。

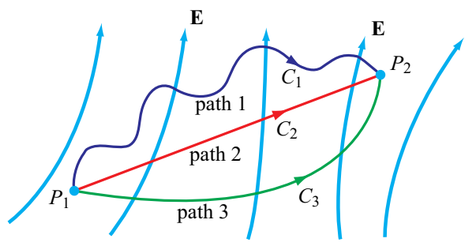


图 4-13 在静电学中，P_2 和 P_1 之间的电势差是相同的，无论用于计算它们之间电场线积分的路径如何。

where V_1 and V_2 are the electric potentials at points P_1 and P_2, respectively. The result of the line integral on the right-hand side of Eq. (4.39) is independent of the specific integration path that connects points P_1 and P_2. This follows immediately from the law of conservation of energy. To illustrate with an example, consider a particle in Earth's gravitational field. If the particle is raised from a height h_1 above Earth's surface to height h_2, the particle gains potential energy in an amount proportional to $(h_2 - h_1)$. If instead we were to first raise the particle from height h_1 to a height h_3 greater than h_2, thereby giving it potential energy proportional to $(h_3 - h_1)$, and then let it drop back to height h_2 by expending an energy amount proportional to $(h_3 - h_2)$, its net gain in potential energy would again be proportional to $(h_2 - h_1)$.

The same principle applies to the electric potential energy W and to the potential difference $(V_2 - V_1)$. The voltage difference between two nodes in an electric circuit has the same value regardless of which path in the circuit we follow between the nodes. Moreover, ***Kirchhoff's voltage law*** states that the net voltage drop around a closed loop is zero. If we go from P_1 to P_2 by path 1 in **Fig. 4-13** and then return from P_2 to P_1 by path 2, the right-hand side of Eq. (4.39) becomes a closed contour, and the left-hand side vanishes. In fact, *the line integral of the electrostatic field* $\mathbf{E}$ *around any closed contour* C *is zero:*

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0. \qquad \textbf{(electrostatics)} \tag{4.40}$$

► A vector field whose line integral along any closed path is zero is called a ***conservative*** or an ***irrotational*** field. Hence, the electrostatic field $\mathbf{E}$ is conservative. ◄

As we will see later in Chapter 6, if $\mathbf{E}$ is a time-varying function, it is no longer conservative, and its line integral along a closed path is not necessarily zero.

The conservative property of the electrostatic field can be deduced from Maxwell's second equation, Eq. (4.1b). If $\partial/\partial t = 0$, then

$$\nabla \times \mathbf{E} = 0. \tag{4.41}$$

If we take the surface integral of $\nabla \times \mathbf{E}$ over an open surface S and then apply Stokes's theorem expressed by Eq. (3.107) to convert the surface integral into a line integral, we obtain

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \tag{4.42}$$

where C is a closed contour surrounding S. Thus, Eq. (4.41) is the differential-form equivalent of Eq. (4.40).

We now define what we mean by the electric potential V at a point in space. Before we do so, however, let us revisit our electric-circuit analogue. Just as a node in a circuit cannot be assigned an absolute voltage, a point in space cannot have an absolute electric potential. The voltage of a node in a circuit is measured relative to that of a conveniently chosen reference point to which we have assigned a voltage of zero, which we call *ground*. The same principle applies to the electric potential V. Usually (but not always), the reference point is chosen to be at infinity. That is, in Eq. (4.39) we assume that $V_1 = 0$ when P_1 is at infinity. Therefore, the electric potential V at any point P is

$$V = -\int_{\infty}^{P} \mathbf{E} \cdot d\mathbf{l} \qquad \text{(V)}. \tag{4.43}$$

Example 4-8: Computing V from $\mathbf{E}$ along Two Paths

A vector field is said to be ***conservative*** if its line integral between two points is the same—irrespective of the path taken between them. In a given region of space, the field $\mathbf{E}$ is given by

$$\mathbf{E} = \hat{\mathbf{x}}\, x^2 + \hat{\mathbf{y}}\, y^2 + \hat{\mathbf{z}}\, z^2. \tag{4.44}$$

(a) Confirm that $\mathbf{E}$ is conservative by demonstrating that $\nabla \times \mathbf{E} = 0$. (b) Compute the potential difference V_{21} between points 1 and 2 in **Fig. 4-14** following the direct path between them. (c) Compute V_{21} by following the path $ABCD$ between points 1 and 2.

Solution: **(a)** The given electric field has components $E_x = x^2$, $E_y = y^2$, and $E_z = z^2$. Applying the curl operator to $\mathbf{E}$ gives

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \hat{\mathbf{x}}\left(\frac{\partial z^2}{\partial y} - \frac{\partial y^2}{\partial z}\right) - \hat{\mathbf{y}}\left(\frac{\partial z^2}{\partial x} - \frac{\partial x^2}{\partial z}\right) + \hat{\mathbf{z}}\left(\frac{\partial y^2}{\partial x} - \frac{\partial x^2}{\partial y}\right)$$

$$= \hat{\mathbf{x}}(0-0) - \hat{\mathbf{y}}(0-0) + \hat{\mathbf{z}}(0-0) = 0.$$

(b) Voltage V_{21} is given by

$$V_{21} = -\int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}.$$

其中 V_1 和 V_2 分别是 P_1 和 P_2 点处的电势。等式右侧的线积分结果： (4.39) 独立于连接点 P_1 和 P_2 的特定积分路径。这直接遵循能量守恒定律。为了举例说明，考虑地球引力场中的一个粒子。如果粒子从地球表面上方的高度 h_1 上升到高度 h_2，粒子将获得与 $(h_2 - h_1)$ 成比例的势能。相反，如果我们首先将粒子从高度 h_1 提升到大于 h_2 的高度 h_3，从而赋予其与 $(h_3 - h_1)$ 成比例的势能，然后通过消耗与 $(h_3 - h_2)$ 成比例的能量让它落回高度 h_2，则其势能的净增益将再次与 $(h_2 - h_1)$。

同样的原理也适用于电势能 W 和电势差 $(V_2 - V_1)$。电路中两个节点之间的电压差具有相同的值，无论我们在节点之间遵循电路中的哪条路径。此外，基尔霍夫电压定律指出，闭环周围的净电压降为零。如果我们通过图 4-13 中的路径 1 从 P_1 到 P_2，然后通过路径 2 从 P_2 返回到 P_1，则方程的右侧为： (4.39) 变成闭合轮廓，左侧消失。事实上，静电场 E 围绕任何闭合轮廓 C 的线积分为零：

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0. \qquad \textbf{(electrostatics)} \tag{4.40}$$

► 沿任何闭合路径的线积分为零的矢量场称为保守场或无旋场。因此，静电场E是保守的。 ◄

正如我们稍后将在第 6 章中看到的，如果 E 是一个时变函数，它就不再是保守的，并且它沿闭合路径的线积分不一定为零。

静电场的保守性质可以从麦克斯韦第二方程（Eq. (4.1b)。如果 $\partial/\partial t = 0$，则

$$\nabla \times \mathbf{E} = 0. \tag{4.41}$$

如果我们在开放曲面 S 上取 $\nabla \times$ E 的曲面积分，然后应用方程 1 表示的斯托克斯定理： (3.107) 将面积分转化为线积分，可得

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \tag{4.42}$$

其中 C 是围绕 S 的闭合轮廓。 (4.41) 是等式的微分形式等价物。 （4.40）。

现在我们定义空间中一点的电势 V 的含义。然而，在此之前，让我们回顾一下我们的电路模拟。正如电路中的节点无法分配绝对电压一样，空间中的点也不可能具有绝对电势。电路中节点的电压是相对于方便选择的参考点的电压来测量的，我们已将零电压分配给该参考点，我们称之为接地。同样的原理也适用于电势 V 。通常（但并非总是），参考点被选择为无穷远。也就是说，在等式中。(4.39) 我们假设当 P_1 无穷大时 $V_1 = 0$。因此，任意点P的电势V为

$$V = -\int_{\infty}^{P} \mathbf{E} \cdot d\mathbf{l} \qquad \text{(V)}. \tag{4.43}$$

示例 4-8：沿着两条路径从 E 计算 V

如果两个点之间的线积分相同，则称向量场是保守的——无论它们之间的路径如何。在给定的空间区域中，场 E 由下式给出

$$\mathbf{E} = \hat{\mathbf{x}}\, x^2 + \hat{\mathbf{y}}\, y^2 + \hat{\mathbf{z}}\, z^2. \tag{4.44}$$

(a) 通过证明 $\nabla \times$ E $= 0$ 来确认 E 是保守的。 (b) 沿着图 4-14 中点 1 和 2 之间的直接路径计算它们之间的电位差 V_{21}。 (c) 沿着点 1 和点 2 之间的路径 ABCD 计算 V_{21}。

解： (a) 给定电场具有分量 $E_x = x^2$、$E_y = y^2$ 和 $E_z = z^2$。将旋度运算符应用于 E 给出

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \hat{\mathbf{x}}\left(\frac{\partial z^2}{\partial y} - \frac{\partial y^2}{\partial z}\right) - \hat{\mathbf{y}}\left(\frac{\partial z^2}{\partial x} - \frac{\partial x^2}{\partial z}\right) + \hat{\mathbf{z}}\left(\frac{\partial y^2}{\partial x} - \frac{\partial x^2}{\partial y}\right)$$

$$= \hat{\mathbf{x}}(0-0) - \hat{\mathbf{y}}(0-0) + \hat{\mathbf{z}}(0-0) = 0.$$

(b) 电压 V_{21} 由下式给出

$$V_{21} = -\int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}.$$

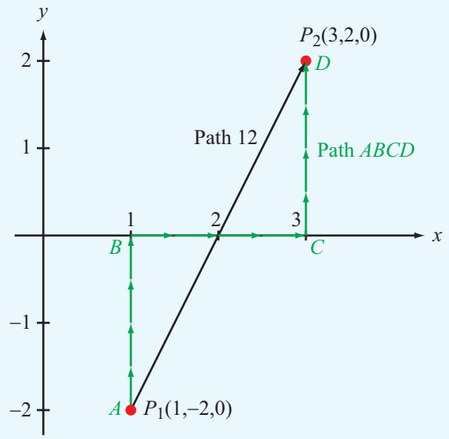


Figure 4-14 Computing V_{21} along two paths (Example 4-8).

The straight-line path resides in the x–y plane, so it is described by the linear form $y = ax + b$. At point 1, $x_1 = 1$ and $y_1 = -2$. Hence,

$$-2 = a + b.$$

Similarly, at point 2, $x_2 = 3$ and $y_2 = 2$, so

$$2 = 3a + b.$$

The two equations lead to $a = 2$, $b = -4$, and

$$y = 2x - 4. \tag{4.45}$$

Since path P_1–P_2 is entirely in the x–y plane, we can set $z = 0$ in the expression for $\mathbf{E}$. Also, we can use the relation given by Eq. (4.45) to reduce $\mathbf{E}$ to a single variable:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{z=0 \text{ and } y=2x-4} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}(2x-4)^2. \tag{4.46}$$

In general,

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz.$$

In the x–y plane, $dz = 0$, and along the straight-line path given by $y = 2x - 4$,

$$dy = 2\,dx. \tag{4.47}$$

Hence,

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,2\,dx. \tag{4.48}$$

The potential difference is then

$$\begin{aligned} V_{21} &= -\int_{P_1}^{P_2} \mathbf{E}\cdot d\mathbf{l} \\ &= -\int_{x=1}^{x=3} [\hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}(2x-4)^2]\cdot[\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,2\,dx] \\ &= -\int_{x=1}^{3} [x^2 + 2(2x-4)^2]\,dx \\ &= -\int_{x=1}^{3} (9x^2 - 32x + 32)\,dx = -14 \quad \text{(V)}. \end{aligned} \tag{4.49}$$

(c) Path $ABCD$ in **Fig. 4-14** consists of three segments.

A* to *B:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{x=1,\ z=0} = \hat{\mathbf{x}}\,1 + \hat{\mathbf{y}}\,y^2, \tag{4.50a}$$

and

$$d\mathbf{l} = \hat{\mathbf{y}}\,dy. \tag{4.50b}$$

B* to *C:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{y=0,\ z=0} = \hat{\mathbf{x}}\,x^2, \tag{4.51a}$$

and

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx. \tag{4.51b}$$

C* to *D:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{x=3,\ z=0} = \hat{\mathbf{x}}\,9 + \hat{\mathbf{y}}\,y^2, \tag{4.52a}$$

and

$$d\mathbf{l} = \hat{\mathbf{y}}\,dy. \tag{4.52b}$$

Hence,

$$\begin{aligned} V_{21} &= -\int_{P_1}^{P_2} \mathbf{E}\cdot d\mathbf{l} \\ &= -\left[\int_{A@x=1,\ y=-2}^{B@x=1,\ y=0} (\hat{\mathbf{x}}\,1 + \hat{\mathbf{y}}\,y^2)\cdot\hat{\mathbf{y}}\,dy + \int_{B@x=1,\ y=0}^{C@x=3,\ y=0} \hat{\mathbf{x}}\,x^2\cdot\hat{\mathbf{x}}\,dx + \int_{C@x=3,\ y=0}^{D@x=3,\ y=2} (\hat{\mathbf{x}}\,9 + \hat{\mathbf{y}}\,y^2)\cdot\hat{\mathbf{y}}\,dy\right] \\ &= -\left[\frac{y^3}{3}\Big|_{-2}^{0} + \frac{x^3}{3}\Big|_{1}^{3} + \frac{y^3}{3}\Big|_{0}^{2}\right] \\ &= -\left[+\frac{8}{3} + \frac{27}{3} - \frac{1}{3} + \frac{8}{3}\right] = -14 \quad \text{(V)}, \end{aligned} \tag{4.53}$$

which is identical with the result given by Eq. (4.49) for the line integral along the straight-line path between points 1 and 2.

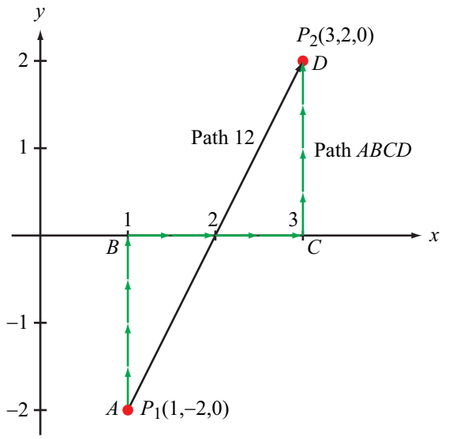


图 4-14 沿两条路径计算 V_{21}（示例 4-8）。

直线路径位于 x-y 平面内，因此由线性形式 y = ax + b 描述。在点 1 处，$x_1 = 1$ 和 $y_1 = -2$。因此，

$$-2 = a + b.$$

类似地，在点 2 处，$x_2 = 3$ 且 $y_2 = 2$，因此

$$2 = 3a + b.$$

这两个方程可得出 a = 2、b = −4，以及

$$y = 2x - 4. \tag{4.45}$$

由于路径 P_1–P_2 完全在 x–y 平面上，因此我们可以在 E 的表达式中将 z = 设置为 0。此外，我们可以使用等式 1 给出的关系。(4.45) 将 E 简化为单个变量：

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{z=0 \text{ and } y=2x-4} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}(2x-4)^2. \tag{4.46}$$

一般来说，

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz.$$

在 x–y 平面中，dz = 0，并沿着 y = 2x − 4 给出的直线路径，

$$dy = 2\,dx. \tag{4.47}$$

因此，

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,2\,dx. \tag{4.48}$$

那么电位差就是

$$\begin{aligned} V_{21} &= -\int_{P_1}^{P_2} \mathbf{E}\cdot d\mathbf{l} \\ &= -\int_{x=1}^{x=3} [\hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}(2x-4)^2]\cdot[\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,2\,dx] \\ &= -\int_{x=1}^{3} [x^2 + 2(2x-4)^2]\,dx \\ &= -\int_{x=1}^{3} (9x^2 - 32x + 32)\,dx = -14 \quad \text{(V)}. \end{aligned} \tag{4.49}$$

(图4-14中的c)路径ABCD由三段组成。

A* to *B:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{x=1,\ z=0} = \hat{\mathbf{x}}\,1 + \hat{\mathbf{y}}\,y^2, \tag{4.50a}$$

和

$$d\mathbf{l} = \hat{\mathbf{y}}\,dy. \tag{4.50b}$$

B* to *C:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{y=0,\ z=0} = \hat{\mathbf{x}}\,x^2, \tag{4.51a}$$

和

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx. \tag{4.51b}$$

C* to *D:

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2\big|_{x=3,\ z=0} = \hat{\mathbf{x}}\,9 + \hat{\mathbf{y}}\,y^2, \tag{4.52a}$$

and

$$d\mathbf{l} = \hat{\mathbf{y}}\,dy. \tag{4.52b}$$

因此，

$$\begin{aligned} V_{21} &= -\int_{P_1}^{P_2} \mathbf{E}\cdot d\mathbf{l} \\ &= -\left[\int_{A@x=1,\ y=-2}^{B@x=1,\ y=0} (\hat{\mathbf{x}}\,1 + \hat{\mathbf{y}}\,y^2)\cdot\hat{\mathbf{y}}\,dy + \int_{B@x=1,\ y=0}^{C@x=3,\ y=0} \hat{\mathbf{x}}\,x^2\cdot\hat{\mathbf{x}}\,dx + \int_{C@x=3,\ y=0}^{D@x=3,\ y=2} (\hat{\mathbf{x}}\,9 + \hat{\mathbf{y}}\,y^2)\cdot\hat{\mathbf{y}}\,dy\right] \\ &= -\left[\frac{y^3}{3}\Big|_{-2}^{0} + \frac{x^3}{3}\Big|_{1}^{3} + \frac{y^3}{3}\Big|_{0}^{2}\right] \\ &= -\left[+\frac{8}{3} + \frac{27}{3} - \frac{1}{3} + \frac{8}{3}\right] = -14 \quad \text{(V)}, \end{aligned} \tag{4.53}$$

这与等式给出的结果相同。(4.49) 为沿点 1 和点 2 之间的直线路径的线积分。